Modern Algebra 1

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I always type my solutions in LaTeX.

Problem 1

Solution. (a) First, we show the reflexive property. For any $a \in S_i$, it follows that $a \in S_i$, and thus $a \sim a$.

Next, we show the symmetric propery. Let $a \sim b$. It follows that $a, b \in S_i$ for some i. Since $a, b \in S_i$, it follows that $b \sim a$.

Finally, we show the transitive propery. Let $a \sim b$ and $b \sim c$. Since $a \sim b$, we have $a, b \in S_i$ for some i. Since $b \sim c$, it follows that $b, c \in S_j$ for some j. For any $S_i \cap S_j \neq \varnothing$ it follows that $S_i = S_j$. Since $b \in S_i, S_j$, it follows that $b \in S_i \cap S_j$ implies $S_i \cap S_j \neq \varnothing$, and thus $S_i = S_j$. Thus, $c \in S_i$. Since $a, c \in S_i$, it follows that $a \sim c$.

(b) Let $a \in S_i$, and consider the set $[a]$. Let $b \in [a]$. It follows that $a \sim b$, and by definition of the equivalence relation $\sim$, we have $b \in S_i$. Therefore, $[a] \subseteq S_i$. Now let $b \in S_i$. By definiton of the relation $\sim$, we have $a \sim b$, and thus $b \in [a]$. Therefore, $[a] = S_i$ for all $a \in S_i$ and for all i. Therefore, the equivalence classes under $\sim$ are precisely the sets S_i.

(c) Let $P = \{g_1 H, \ldots, g_n H\}$ be the set of all the cosets of H in G. By the theorem concerning properties of cosets, for any two cosets $g_i H, g_j H$, we have either $g_i H \cap g_j H = \varnothing$ or $g_i H = g_j H$. It remains only to be shown that $\bigcup_{i=1}^{n} g_n H = G$.

Now suppose for purpose of contradiction that there exists some $a \in G$ such that for all i, $a \notin g_i H$. Since H is a subgroup, $e \in H$, and thus $a = ae \in aH$. By the one theorem concerning properties of cosets, we know that since $a \notin g_i H$ for all H, then $aH \neq g_i H$ for all i, and thus P is not the set of all cosets, since we have identified a new distinct coset $aH \notin P$. By contradiction, we have $a \in g_i H$ for some i for all a. Therefore, P is a partition of G. ■

Problem 2

Solution. Let L and R be the sets of left and right cosets, respectively, of H in G. Let $\phi : L \to R$ be the map $gH \mapsto Hg^{-1}$. First, we show the map is well defined. Let $aH = bH$. It follow that for all $h_1 \in H$, there exists some $h_2 \in H$ such that $ah_1 = bh_2$. It follows that

$$\begin{aligned}
& (ah_1)\left(h_1^{-1}a^{-1}\right) = (bh_2)\left(h_1^{-1}a^{-1}\right) \\
\implies & e = (bh_2)\left(h_1^{-1}a^{-1}\right) \\
\implies & h_2^{-1}b^{-1} = \left(h_2^{-1}b^{-1}\right)(bh_2)\left(h_1^{-1}a^{-1}\right) \\
\implies & h_2^{-1}b^{-1} = h_1^{-1}a^{-1}.
\end{aligned}$$

We know from the properties of cosets that $a^{-1} \in Hb^{-1}$ if and only if $Ha^{-1} = Hb^{-1}$. We have

$$\begin{aligned}
& h_2^{-1}b^{-1} = h_1^{-1}a^{-1} \\
\implies & h_1 h_2^{-1}b^{-1} = a^{-1}.
\end{aligned}$$

Since $h_1 h_2^{-1} \in H$, we have $h_1 h_2^{-1} b^{-1} \in Hb^{-1}$, and thus $a \in Hb^{-1}$. Therefore, if $aH = bH$, then $Ha^{-1} = Hb^{-1}$, and the map is well defined.

Now, we show the map is surjective. Let $a \in G$ and $Ha \in R$. It follows that

$$Ha = H\left(a^{-1}\right)^{-1} = \phi\left(a^{-1}H\right).$$

Therefore, ϕ is surjective. Now we show the map is injective. Suppose there exist $aH, bH \in L$ such that $\phi(aH) = \phi(bH)$. It follows that $Ha^{-1} = Hb^{-1}$, and thus for all $h_1 \in H$, there exists $h_2 \in H$ such that $h_1 a^{-1} = h_2 b^{-1}$. It follows that

$$\begin{aligned} e &= \left(h_2 b^{-1}\right)\left(a h_1^{-1}\right) \\ \implies b h_2^{-1} &= a h_1^{-1} \\ \implies b h_2^{-1} h_1 &= a. \end{aligned}$$

Since $h_2^{-1} h_1 \in H$, then $a \in bH$, and thus $aH = bH$. Therefore, ϕ is bijective, and $|L| = |R|$. ■

Problem 3

Solution. Define $\phi : G \to G$ as the map $a \mapsto a^n$. Let $a, b \in G$. We have

$$\phi(ab) = (ab)^n.$$

Since G is abelian,

$$(ab)^n = a^n b^n = \phi(a)\phi(b).$$

Therefore, ϕ is a homomorphism. Now, we show ϕ is surjective. Let $b \in G$. Since n and $|G|$ are relatively prime, there exist $\alpha, \beta \in \mathbb{Z}$ such that $\alpha n + \beta |G| = 1$. We then have

$$g = g^1 = g^{\alpha n + \beta |G|} = g^{\alpha n} g^{\beta |G|}.$$

By Lagrange's theorem, $|g|$ divides $|G|$. Therefore, there exists some $\ell \in \mathbb{Z}$ such that $|G| = \ell |g|$. By definition, $g^{|g|} = e$. Therefore,

$$\begin{aligned} g^{\alpha n} g^{\beta |G|} &= g^{\alpha n} g^{\beta \ell |g|} \\ &= g^{\alpha n} \left(g^{|g|}\right)^{\beta \ell} \\ &= g^{\alpha n} e^{\beta \ell} \\ &= g^{\alpha n}. \end{aligned}$$

Therefore, $g = g^{\alpha n} = (g^{\alpha})^n$ for some α. Define $a \coloneqq g^{\alpha}$. Then $a^n = g$. Therefore, ϕ is surjective.

Finally, we show ϕ is injective. I did this in a really weird way, and after the fact a friend showed me the more straightforward method where $\left|ab^{-1}\right|$ divides both n and $|G|$. Since I didn't come up with that on my own, I haven't included it, but I think my method is still correct. It's just really excessive.

Let $a, b \in G$, and suppose $\phi(a) = \phi(b)$. First, we will show $\phi^k(a) = a^{n^k}$. We will use the method of proof by induction. Clearly, we have the base case of $\phi^1(a) = a^n = a^{n^1}$. Now suppose $\phi^k(a) = a^{n^k}$. We wish to show $\phi^{k+1}(a) = a^{n^{k+1}}$. It follows that

$$\phi^{k+1}(a) = \phi\left(\phi^k(a)\right) = \phi\left(a^{n^k}\right) = \left(a^{n^k}\right)^n = a^{n\left(n^k\right)} = a^{n^{k+1}}.$$

By induction, $\phi^k = a^{n^k}$.

Next, we will show that if $\phi(a) = \phi(b)$, then $\phi^k(a) = \phi^k(b)$ for any $k \in \mathbb{N}$. We will again use the method of induction, but we assume our base case of $k = 1$ is true in this proof, so we need only show the second step of induction. Suppose $\phi^k(a) = \phi^k(b)$ for some k. We wish to show $\phi^{k+1}(a) = \phi^{k+1}(b)$. Notice that since $\phi^k(a) = \phi^k(b)$, and since inverses are unique, we have $\left(\phi^k(a)\right)^{-1} = \left(\phi^k(b)\right)^{-1}$. It follows that

$$\begin{aligned}
& \left(\phi^k(a)\right)^n = \left(\phi^k(b)\right)^n \\
\iff & \left(\phi^k(a)\right)^n \left(\left(\phi^k(a)\right)^{-1}\right)^n = \left(\phi^k(b)\right)^n \left(\left(\phi^k(a)\right)^{-1}\right)^n \\
\iff & \left(\phi^k(a)\right)^n \left(\left(\phi^k(a)\right)^n\right)^{-1} = \left(\phi^k(b)\right)^n \left(\left(\phi^k(b)\right)^n\right)^{-1} \\
\iff & e = e.
\end{aligned}$$

Since $e = e$, and since all statements were if and only if statements, it follows that $\phi^k(a) = \phi^k(b)$ for all k.

Suppose there exists some k such that $n^k \equiv 1 \pmod{|G|}$. It follows that there exists some $q \in \mathbb{Z}$ such that $n^k = q\,|G| + 1$. Since for any $a \in G$, the order of a divides $|G|$, it follows that

$$a^{n^k} = a^{q|G|+1} = a^{q|G|}a = ea = a.$$

It remains to be shown such a k exists.

Since n is relatively prime to $|G|$, then $n \in U(|G|)$. Note that $|U(|G|)| = \varphi(|G|)$, where φ is the Euler totient function. By Lagrange's Theorem, $|n|$ divides $|U(|G|)|$, and thus $n^{|U(|G|)|} \equiv n^{\varphi(|G|)} \equiv 1 \pmod{|G|}$, since 1 is the identity in $U(|G|)$. It follows that

$$\begin{aligned}
& \phi^{\varphi(|G|)}(a) = \phi^{\varphi(|G|)}(b) \\
\implies & a^{n^{\varphi(|G|)}} = b^{n^{\varphi(|G|)}} \\
\implies & a = b.
\end{aligned}$$

Since $a = b$, ϕ is injective, and is thus an automorphism on any finite abelian group G. ■

Problem 4

Solution. Let $a \in G$ such that $a \in H$ and $a \in K$. By Lagrange's theorem, $|a|$ divides both $|H|$ and $|K|$. Since H and K have relatively prime orders, the order of a must be no greater than 1, and thus can only be 1. Therefore, $a = e$, and $H \cap K = \{e\}$. ■

Problem 5

Solution. Let G be a group with $|G| = 4$. By Lagrange's theorem, for all $a \in G$, the order of a must divide 4. Suppose there exists some $a \in G$ with $|a| = 4$. Note that $a^1 = a$ and $a^4 = e$. We know that $a^2 \neq e$ and $a^3 \neq e$, since $2, 3 < 4$ and $|a| = 4$. Thus, if we can show that $a^2 \neq a$, $a^3 \neq a$, and $a^2 \neq a^3$, we will have shown that there are 4 distinct elements of G that can be represented as powers of a. Since there are only 4 elements of G, this implies $\langle a \rangle = G$. We will show all three statements by contradiction.

$$\begin{aligned}
a^2 = a &\implies a = e \implies |a| = 1 \Longrightarrow\Longleftarrow \\
a^3 = a^2 &\implies a = e \implies |a| = 1 \Longrightarrow\Longleftarrow \\
a^3 = a &\implies a^2 = e \implies |a| = 2 \Longrightarrow\Longleftarrow .
\end{aligned}$$

Therefore, if there exists $a \in G$ such that $|a| = 4$, then $G = \langle a \rangle$. Let $b, c \in G$. Since G is generated by a, there exist integers k, ℓ such that $b = a^k$ and $c = a^\ell$. It follows that

$$bc = a^k a^\ell = a^{k+\ell} = a^{\ell+k} = a^\ell a^k = cb.$$

Therefore, G is abelian.

Now suppose there does not exist $a \in G$ such that $|a| = 4$. Notice that by contrapositive, G is not cyclic. By Lagrange's theorem, for any $a \in G$, the order of a divides 4. The only positive divisors of 4 are 1, 2, and 4. The only element with order 1 is the identity e, so G must consist of precisely e and three elements of order 2. For any $a \in G$, since either $|a| = 2$ or $a = e$, we have $aa = e$, and thus $a = a^{-1}$. Let $a, b \in G$. Notice that $ab \in G$, and thus $(ab)^{-1} = ab$. Additionally, we know $(ab)^{-1} = b^{-1}a^{-1}$. We thus have $ab = b^{-1}a^{-1}$. Recall that $b^{-1} = b$ and $a^{-1} = a$. It follows that $ab = ba$, and G is abelian. ∎